

fp-tools: A Reproducible Platform for ATAC-seq Footprinting and Regulatory Motif Analysis

Yaoxiang Li ¹ and Chunling Yi ^{1,*}

¹ Department of Oncology, Lombardi Comprehensive Cancer Center, Georgetown University Medical Center, Georgetown University, Washington, DC, USA; yl814@georgetown.edu (Y.L.); cy232@georgetown.edu (C.Y.)

* Correspondence: cy232@georgetown.edu (C.Y.)

Abstract

ATAC-seq footprinting enables near-base-resolution inference of transcription-factor (TF) occupancy across the genome. However, the field lacks a computationally efficient, user-friendly, and integrated platform that accepts both bulk and single-cell ATAC-seq data, supports both de novo motif discovery and matching to public motif databases, and generates interactive and publication-ready visual outputs. Here, we present *fp-tools*, a Python package that builds on the widely used ATAC-seq footprinting framework TOBIAS. We substantially improve computational efficiency across key steps, including Tn5 bias correction, footprint calling, and motif matching. In addition, *fp-tools* introduces de novo motif discovery, replicate-aware differential footprinting, and scaled motif aggregation, all features not provided by TOBIAS. We further streamline a single-cell ATAC-seq footprinting workflow, provide interactive visualization of analysis results, and implement a graphical user interface (GUI) that enables code-free execution of all package commands. Using public bulk ATAC-seq replicates and single-cell datasets, we demonstrate that *fp-tools* quantitatively recovers cell-type-specific TF-binding patterns and identifies de novo binding motifs that are not captured by existing public motif databases. Together, *fp-tools* provides a highly efficient, accessible, and comprehensive platform for TF footprint analysis across bulk and single-cell ATAC-seq studies.

Keywords: ATAC-seq; transcription factor footprinting; chromatin accessibility; motif analysis; replicate-aware differential footprinting; de novo motif discovery

1. Introduction

Assay for Transposase-Accessible Chromatin using Sequencing (ATAC-seq) has become a standard tool in epigenetic and transcriptional studies [1]. By using a hyperactive Tn5 transposase to insert sequencing adapters into accessible regions of DNA, ATAC-seq enables rapid and sensitive identification of regulatory elements such as promoters, enhancers, and insulators. With the development of single-cell ATAC-seq (scATAC-seq), this approach has further enabled the characterization of cell-type-specific regulatory programs within complex tissues [2].

Less widely recognized is that deeper analysis of the local depletion of Tn5 cut sites within accessible chromatin can reveal transcription factor (TF) occupancy at near base-pair resolution, producing characteristic protected regions known as “footprints.” Compared with standard motif enrichment analysis, which evaluates only the frequencies of TF motifs within accessible regions, footprinting provides more direct evidence of TF binding and quantitative information about binding strength by measuring the difference in accessibility

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between the protected footprint center and its flanking regions. However, the requirement for additional computational processing, careful correction for Tn5 insertion bias, and specialized analytical tools has limited its broader adoption.

Earlier footprinting methods, including CENTIPEDE, Wellington, PIQ, DeFCoM, and HINT, established important frameworks for detecting TF occupancy from chromatin accessibility data [3–7]. Several of these methods were developed primarily around DNase-seq and did not directly address the strong sequence preferences of Tn5 transposase in ATAC-seq. HINT-ATAC incorporated Tn5 bias into footprint detection by modeling position-dependent cleavage patterns [7], whereas TOBIAS further refined Tn5 correction by first estimating expected insertion bias from local sequence context and adjusting the observed cut-site signal before footprint scoring [8]. This explicit correction reduces the likelihood that sequence-driven Tn5 insertion preferences are misinterpreted as TF-associated protection, thereby improving the reliability of ATAC-seq footprint inference.

More recently, PRINT/scPrinter, maxATAC, and ChromBPNet have advanced ATAC-seq-based TF binding inference through multiscale footprinting and deep-learning models [9–11]. Although powerful, these approaches remain challenging for broad adoption by experimental researchers because they often require complex software setup, large training or reference datasets, GPU-oriented computation, or less transparent model interpretation. In contrast, TOBIAS provides a more accessible ATAC-seq footprinting framework with explicit Tn5 bias correction and interpretable footprint scores [8]. However, TOBIAS remains primarily command-line based and does not natively support *de novo* motif discovery, replicate-aware statistics, or streamlined single-cell ATAC-seq footprinting workflows.

We developed *fp-tools* to address these unmet needs. The package preserves the interpretable, motif-centered footprinting logic of TOBIAS while improving the speed, usability, and scope of the core analysis. Compared with TOBIAS, *fp-tools* adds *de novo* motif discovery with database matching, replicate-aware differential footprint analysis, scaled motif aggregation, streamlined single-cell ATAC-seq footprinting, interactive HTML reports, editable SVG outputs, and a graphical user interface (GUI) for code-free execution of command-compatible workflows.

By connecting common TF-footprinting tasks into a unified, auditable workflow that can be run from the command line, saved YAML configuration files, or the GUI, *fp-tools* makes footprinting more accessible to both computational and experimental researchers. Here, we describe the implementation of *fp-tools* and evaluate its performance using public bulk ATAC-seq replicate datasets and single-cell ATAC-seq datasets, demonstrating its potential to expand the routine use of footprinting in regulatory genomics studies.

2. Materials and Methods

2.1. Core Signal Model and Footprint Scoring

fp-tools uses a command-oriented workflow in which corrected cut-site tracks, genomic regions, motif annotations, sample labels, and output files follow consistent conventions. This allows bias-correction outputs to be used directly for footprint scoring, motif-anchored differential analysis, and aggregate visualization around transcription factor binding sites (TFBSs) or user-defined genomic regions. YAML configuration files record input files and analysis parameters so the same workflow can be rerun without modifying the underlying methods.

Following the signal model popularized by TOBIAS [8], *fp-tools* treats base-resolution, bias-corrected Tn5 cut-site profiles as the primary signal for footprint analysis. This design preserves the standard bulk ATAC-seq footprinting interpretation, in which putative TF occupancy is inferred from a local depletion of corrected cut sites at a motif or candidate binding site relative to the surrounding accessible flanks. Bias correction

estimates the sequence-driven component of Tn5 insertion and produces corrected cut-site tracks in which local depletion and enrichment are less likely to reflect transposase sequence preference alone. After bias correction, `fp-tools` can optionally rescale corrected cut-site tracks before footprint scoring. When this option is enabled, a single multiplicative scale factor is estimated for each sample from shared background regions. If $q_{s,0.95}$ denotes the 95th percentile of corrected signal for sample s over the shared background regions and $\tilde{q}_{0.95}$ denotes the across-sample target, typically the median of the sample-level 95th percentiles, the rescaled signal is

$$x'_{s,i} = x_{s,i} \frac{\tilde{q}_{0.95}}{q_{s,0.95}}.$$

This robust q95 scaling reduces sample-to-sample differences in overall signal magnitude while preserving the shape of each sample's corrected signal profile, without forcing the complete empirical signal distributions to be identical.

For footprint scoring, let x_i denote the corrected Tn5 cut-site signal at position i around a candidate footprint position p , after optional q95 scaling when enabled. The default footprint score is calculated as a center-versus-flank contrast,

$$F(p) = \frac{1}{|L| + |R|} \left(\sum_{i \in L} x_i + \sum_{i \in R} x_i \right) - \frac{1}{|C|} \sum_{i \in C} x_i,$$

where C denotes the protected center window and L and R denote the left and right flanking windows. Window sizes are user-configurable and are recorded with the analysis parameters. Larger values indicate lower corrected cut-site signal in the center than in the surrounding flanks, corresponding to the expected footprint pattern of reduced Tn5 cleavage at a putative bound site and higher cleavage in nearby accessible DNA.

2.2. Known-motif Scanning and Motif-site Definition

For motif-anchored footprint analysis, `fp-tools` scans candidate accessible regions for transcription factor motif matches using position weight matrices from a user-supplied motif database, such as JASPAR 2026 CORE vertebrates non-redundant motifs [12]. The GC content of the accessible-region sequence set is used as the background nucleotide distribution for log-odds motif scoring, and motif-specific score thresholds are derived from the configured motif-match p-value cutoff. Both strands are scanned. For each motif, overlapping candidate matches for that motif are resolved by retaining the higher-scoring match, so each motif contributes a non-overlapping site set for downstream analysis.

Each motif site is represented by its genomic interval, motif identifier, strand, motif score, associated peak or region annotation, and the footprint score measured from each input signal track. For motif m , let S_m denote the candidate sites assigned to that motif within the common accessible-region universe. Background positions B are sampled from the same accessible-region universe and are used to estimate the expected distribution of footprint-score changes not specifically associated with scanned motif sites. This shared motif-site and background-position representation allows single-sample, replicate-supported, and de novo-augmented analyses to use the same downstream differential footprinting framework.

When de novo motifs are discovered and matched to known motifs, the discovered motif set can be analyzed independently or appended to the known motif database and scanned using the same motif-site conventions.

2.3. De Novo Motif Discovery

The de novo motif discovery module begins from candidate footprint intervals, which may be produced by footprint calling or provided as input. For each candidate interval j with center p_j , BED-like strand annotation a_j , and flank size h , it extracts a candidate-centered sequence from the reference genome. Using one-based inclusive notation for clarity, this sequence has length $2h + 1$ and is written as

$$S_j = \begin{cases} G[p_j - h, p_j + h], & a_j \in \{+, \cdot\}, \\ \text{revcomp}\{G[p_j - h, p_j + h]\}, & a_j = -. \end{cases}$$

Thus, positive-strand and unstranded candidate intervals are exported in the reference-genome orientation, whereas negative-strand intervals are reverse-complemented before motif discovery. When half-open BED-style coordinates are used internally, interval endpoints are converted before sequence extraction so that the exported FASTA sequence remains centered on p_j . The module records the input intervals, genome build, flank width, software versions, and the parameters passed to MEME, DREME, or STREME [13–15] for motif discovery. When database comparison is requested, the corresponding Tomtom [16] settings are also recorded. STREME was designed for accurate motif discovery from enriched sequence sets [15]. Discovered motifs can be analyzed as an independent motif set or matched to a known database such as JASPAR 2026 [12] and appended to the known motif set for downstream differential footprint analysis. In this design, candidate FASTA export, motif discovery, motif comparison, and database augmentation become part of the recorded footprint analysis.

2.4. Pseudobulk Fragment Aggregation

The pseudobulk module groups single-cell ATAC fragments by a metadata column, such as cell type, cluster, donor, or treatment. Let F_b be the multiset of fragments assigned to barcode b , and let $g(b)$ be its group label. The pseudobulk fragment collection for group k is the concatenated multiset

$$F_k = \text{concat}_{b:g(b)=k}(F_b),$$

so repeated fragments and fragment multiplicity are preserved. Thus, the pseudobulk profile represents the sum of fragments from cells assigned to a group, rather than a binarized representation of accessible regions. Groups that pass specified cell-count and fragment-count filters are written as bulk-like fragment files with manifest records. When genome sizes are supplied, `fp-tools` can also emit counts per million cut sites (CPM)-normalized cut-site bigWigs,

$$y_{k,t} = 10^6 \frac{c_{k,t}}{\sum_u c_{k,u}},$$

where $c_{k,t}$ is the cut-site count for group k at genomic position t . These tracks support motif-centered aggregate profiles matched to the bulk ATAC-seq analysis path. The full pseudobulk route writes pseudo-paired BAMs from grouped fragment endpoints and then runs correction, scoring, motif-aware comparison, and plotting with the same conventions used for bulk ATAC-seq.

2.5. Cell-level Footprint-signature Visualization

For cell-level marker visualization, `fp-tools` computes K-nearest-neighbor (KNN)-smoothed footprint signatures from single-cell fragments. For each marker TF, motif centers are scanned inside the selected accessible-bin universe. Fragment cut sites are counted in a ± 100 bp window around those motif centers for each barcode. Each cell is assigned a

local neighborhood profile by summing cut-site profiles across its 75 nearest neighbors in the single-cell embedding, and a flank-minus-center footprint-signature score is computed from that neighborhood profile. Marker TFs are selected from the pseudobulk differential-footprint results and biological prior marker sets before display orientation is applied. For visualization only, signatures are oriented so that higher values correspond to the marker-associated cell population and are then standardized per TF for UMAP display. These KNN-smoothed signatures are used for cell-level visualization of motif-centered cut-site patterns, whereas full bias-corrected footprint tracks and statistical comparisons are computed from grouped pseudobulk profiles.

2.6. Replicate-aware Differential Footprint Analysis

The differential footprint module supports both single-sample contrasts and replicate-supported comparisons. It retains conventional motif-centered footprint-score logic while adding repeated condition labels, replicate-level summaries, and replicate-aware testing. Repeated condition names, such as `-cond-names Bcell Bcell Tcell Tcell`, define biological replicate groups while preserving the original differential-footprint output structure. Let $x_{c,r,i}$ be the footprint score at site or background position i for replicate r of condition c .

For single-sample versus single-sample contrasts, `diff-footprints` uses the motif-site versus background-distribution comparison of the standard TOBIAS-style report. Candidate sites are grouped by motif, site-level changes at motif sites are compared with changes at sampled background positions, and p-values are adjusted across motifs within each contrast.

For replicate-supported comparisons, `fp-tools` generalizes this comparison to multiple samples per condition. Footprint scores are computed for each replicate on the same peak and motif-site universe. Replicate scores are then summarized within each condition,

$$\bar{x}_{c,i} = \frac{1}{n_c} \sum_{r=1}^{n_c} x_{c,r,i}, \quad s_{c,i}^2 = \frac{1}{n_c - 1} \sum_{r=1}^{n_c} (x_{c,r,i} - \bar{x}_{c,i})^2.$$

For motif m , let S_m denote its candidate motif sites and let B denote the sampled background positions from the same accessible-region universe. The replicate-aware motif-background effect is reported as

$$\Delta_m = \frac{1}{|S_m|} \sum_{i \in S_m} (\bar{x}_{c_1,i} - \bar{x}_{c_2,i}) - \frac{1}{|B|} \sum_{i \in B} (\bar{x}_{c_1,i} - \bar{x}_{c_2,i}).$$

The report also includes replicate counts, condition means, SDs, and SE summaries, so the additional samples contribute to a more stable estimate rather than being collapsed outside the analysis.

Quantile-based score scaling remains an optional comparison-stage adjustment. In single-sample contrasts, sample-level quantile scaling provides the conventional two-sample score-distribution matching. In replicate-supported comparisons, condition-level background quantile scaling fits a quantile-based normalization curve from positive background-score quantiles for each condition and applies the corresponding condition curve before condition summaries are computed. This option is used as a sensitivity adjustment; the main replicate-aware extension is the explicit use of replicate-level motif scores.

When per-sample motif-score matrices are available, `fp-tools` further applies empirical-Bayes variance moderation across motifs, following the moderated testing framework of Smyth [17]. This replicate-supported matrix view estimates condition effects from

per-sample motif-site scores and is reported alongside, rather than replacing, the native motif-background columns. If

$$z_{c,r,m} = \frac{1}{|S_m|} \sum_{i \in S_m} x_{c,r,i}$$

is the per-sample motif-site score, the contrast is

$$\hat{\beta}_m = \frac{1}{n_{c_1}} \sum_r z_{c_1,r,m} - \frac{1}{n_{c_2}} \sum_r z_{c_2,r,m}, \quad t_m^{EB} = \frac{\hat{\beta}_m}{\sqrt{s_{m,EB}^2 (n_{c_1}^{-1} + n_{c_2}^{-1})}},$$

where $s_{m,EB}^2$ is the variance moderated across motifs. P-values from this matrix view are adjusted across motifs within each contrast using Benjamini–Hochberg correction and reported alongside the native motif-background statistics. The native report preserves the motif-versus-background effect Δ_m , whereas the matrix view provides a replicate-supported audit of motif-site score changes across biological samples.

2.7. Public Data Processing and Evaluation

The bulk replicate demonstration uses a 3-vs-3 public ENCODE ATAC-seq comparison between K562 and HepG2, with three released GRCh38 alignment BAMs per cell line. Problematic genomic regions were filtered using the ENCODE blacklist [18]. Replicate peak intervals were merged into a common peak universe, bias-corrected, q95-scaled over shared background regions, footprint-called, and analyzed by replicate-aware differential footprint comparison. The analysis used JASPAR 2026 CORE vertebrates non-redundant motifs [12] for known-motif scanning. Exact command-line parameters and software versions are recorded in source TSV files in the manuscript tables directory and are summarized in Tables S1 and S2.

Public inputs are organized as versioned TSV manifests that record source URLs, accessions, assemblies, checksums, local paths, and analysis labels. Discovery and download scripts query public repositories such as ENCODE [19] for released human GRCh38 ATAC-seq records and write resumable download reports. The manuscript evaluation combines compact software-regression test inputs, public bulk ATAC replicates, de novo motif discovery examples, and public PBMC single-cell ATAC examples. For each analysis, command-line parameters, software versions, resolved input manifests, source tables, and generated figure inputs are kept with the manuscript resources so the reported examples can be rerun and audited.

2.8. Software Testing and Reproducibility

The package includes unit tests for numerical helpers and CLI behavior, plus automated output checks for footprint calling, differential footprint analysis, and aggregate plotting, with an opt-in slow check for bias correction. Continuous integration runs the suite on compact reference inputs, builds source and wheel artifacts, and verifies the installed console scripts. The repository also includes `environment.yml`, a Dockerfile, release metadata, and Make targets for testing, building, and manuscript checks. These choices follow FAIR-oriented research-software principles and established bioinformatics environment practices [20–22]. Analysis scripts record random seeds, tool versions, command lines, and resolved manifests alongside the corresponding outputs. Figure source tables and editable figure files are retained with the manuscript resources for auditability.

3. Results

We evaluated `fp-tools` using public bulk and single-cell ATAC-seq datasets to assess workflow reproducibility, biological interpretability, and practical usability. We first examined the integrated footprinting workflow, then evaluated de novo motif discovery, replicate-aware differential footprint analysis, pseudobulk single-cell ATAC footprinting, and the reproducibility, runtime, and graphical review features of the package.

3.1. A Unified, Reproducible Footprinting Platform

`fp-tools` consolidates the major steps of ATAC-seq footprint analysis into a command-compatible workflow. Bias correction, footprint calling, known-motif scanning, de novo motif discovery, differential footprint analysis, aggregate visualization, and pseudobulk single-cell analysis share consistent conventions for genomic regions, signal tracks, motif annotations, sample labels, and output files. This reduces the need for manual file conversion or custom bridging scripts between analysis steps. YAML configuration files record the same analysis in a machine-readable form, allowing workflows to be rerun, reviewed, and adapted for batch processing (Figure 1).

This design keeps the interpretable TOBIAS-style footprinting model, in which bias-corrected cut-site signal is summarized by motif-centered center-versus-flank depletion, while extending the workflow to include de novo motif discovery, replicate-aware reporting, pseudobulk single-cell analysis, interactive HTML reports, editable figure outputs, and a browser-based GUI.

3.2. De Novo Motif Discovery Complements Database-based Motif Analysis

The de novo motif discovery path converts candidate footprint intervals into centered FASTA files, runs MEME, DREME, or STREME, optionally compares discovered motifs with a known database using Tomtom, and writes TSV/HTML summaries with inline consensus logos. This extends the TOBIAS-style known-motif path by adding a recorded handoff from footprint candidates to motif discovery; it does not change the center-versus-flank footprint scoring model. The module can be run alone to discover recurring sequence patterns in candidate intervals, or after known-motif scanning to identify motifs that are absent from the selected database.

We used public ENCODE K562 and HepG2 ATAC-seq data as two examples to evaluate whether candidate footprint intervals contain recurring sequence motifs beyond a curated database. Candidate footprints were called from bias-corrected cut-site tracks, candidate FASTA files were exported with 75 bp flanks, and reciprocal STREME runs were performed using one cell line as the enriched sequence set and the other as the sequence-control set. Discovered motifs were then summarized, compared with JASPAR by Tomtom, and scanned back into candidate intervals for aggregate footprint review.

The known-motif scan used JASPAR 2026 vertebrate non-redundant motifs, and reciprocal STREME runs contributed candidate-derived motifs from both ENCODE cell lines. Figure 2B summarizes the merged de novo motif-supported windows as either shared with the JASPAR-supported scan or detected only by the de novo scan. The shared component dominated both K562 and HepG2, while de novo-only windows accounted for 6.0% of the total de novo-supported merged window base pairs.

Representative discovered motifs showed clear motif-centered aggregate footprint profiles in K562 and HepG2 (Figure 2C,D), indicating that the de novo sequence patterns correspond to footprinted cut-site structure in the candidate intervals. Together with the recorded FASTA export, motif discovery, and motif-comparison workflow (Figure 2A), these results support using de novo motif discovery as a complement to database-based motif matching when the chosen motif collection is incomplete or conservative.

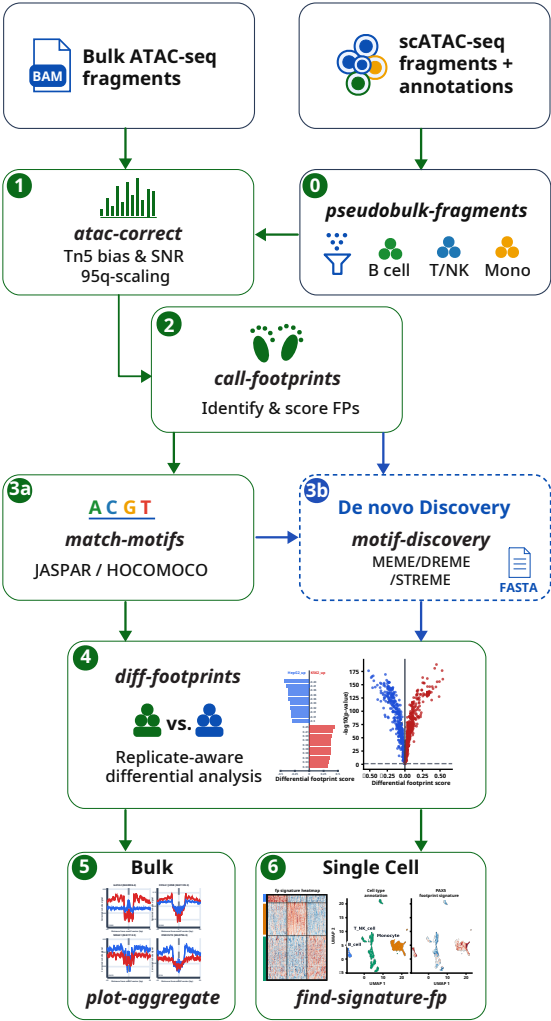


Figure 1. Overview of the fp-tools ATAC-seq footprinting platform. Bulk ATAC-seq BAM/fragments and single-cell ATAC fragments with annotations enter a shared command-oriented analysis. The analysis supports Tn5 bias correction, footprint calling, motif matching against curated databases, de novo motif discovery, replicate-aware differential footprint analysis, aggregate footprint visualization, and pseudobulk single-cell footprint analysis.

3.3. Replicate-aware Differential Footprint Analysis

We next evaluated replicate-aware differential footprint analysis using a 3-vs-3 public ENCODE ATAC-seq comparison between K562 and HepG2. All six replicates were scanned on the same merged peak and motif-site universe before comparison, so condition summaries, replicate-level variation, and motif rankings were reported in a common coordinate framework.

A representative K562 versus HepG2 differential footprint report is shown in Figure 3. In this 3-vs-3 comparison, the report ranked HNF4A, ONECUT2, and HNF4G motifs among the HepG2-biased signatures and GATA2, TRPS1, and SP9 motifs among the K562-biased signatures. The bar plot and volcano panel show the same directional separation, with HepG2-biased motifs on the negative side of the differential footprint-score axis and K562-biased motifs on the positive side. The replicate-level aggregate profiles provide a second check on these rankings: for each selected motif, the three K562 and three HepG2 traces are displayed separately, making replicate consistency and motif-centered cut-site structure visible in the same report.

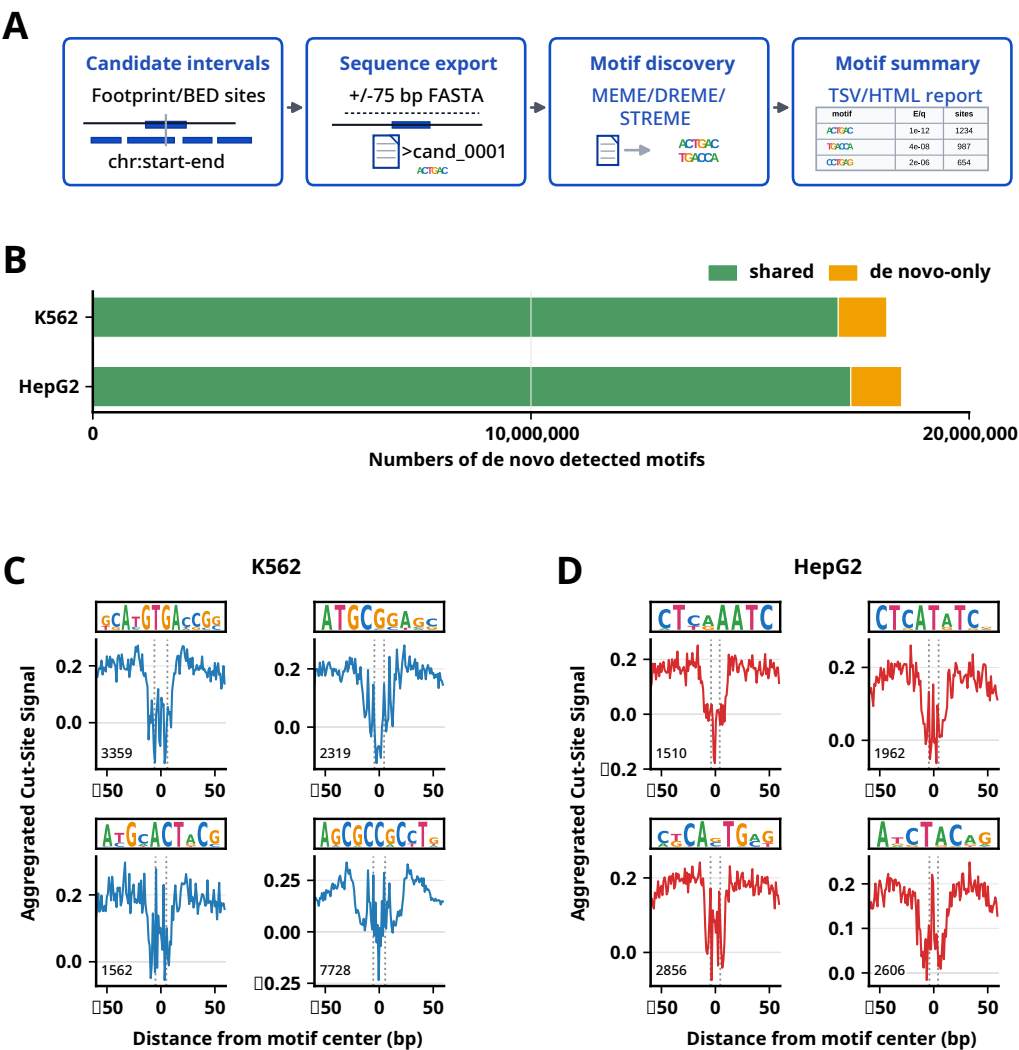


Figure 2. De novo motif discovery and validation summary. (A) Candidate footprint or BED intervals are converted into centered FASTA sequences, analyzed with MEME/DREME/STREME, and summarized as motif tables and reports. (B) Merged site-window coverage compares shared and de novo-only motif-supported intervals in K562 and HepG2. (C,D) Representative K562 and HepG2 de novo motifs show footprint-like normalized cut-site aggregate profiles centered on motif instances.

3.4. Single-cell ATAC Fragments Support Pseudobulk Footprinting

We evaluated pseudobulk aggregation using the public 10x Genomics 5k PBMC single-cell ATAC dataset [23]. The demonstration used the Cell Ranger ATAC 2.0.0 fragment file for atac_pbmc_5k_nextgem and broad cell annotations from the PBMC5k single-cell embedding. After filtering, 4,437 annotated cells were grouped into B cells (422 cells), monocytes (1,702 cells), and T/NK cells (2,313 cells). All three groups passed the pseudobulk filters and contained 2.1, 10.8, and 13.3 million counted fragments, respectively.

The marker-by-cell heatmap in Figure 4A shows that KNN-smoothed footprint-signature scores separate the three broad PBMC groups. B-cell-associated rows include STAT6 and FOSB; monocyte-associated rows include CEBPA, IRF8, and RELA; and T/NK-associated rows include ZNF683, NR4A1, and SMAD3. These row-level patterns show that motif-centered footprint signatures derived from grouped single-cell fragments remain visible at the cell-label level.

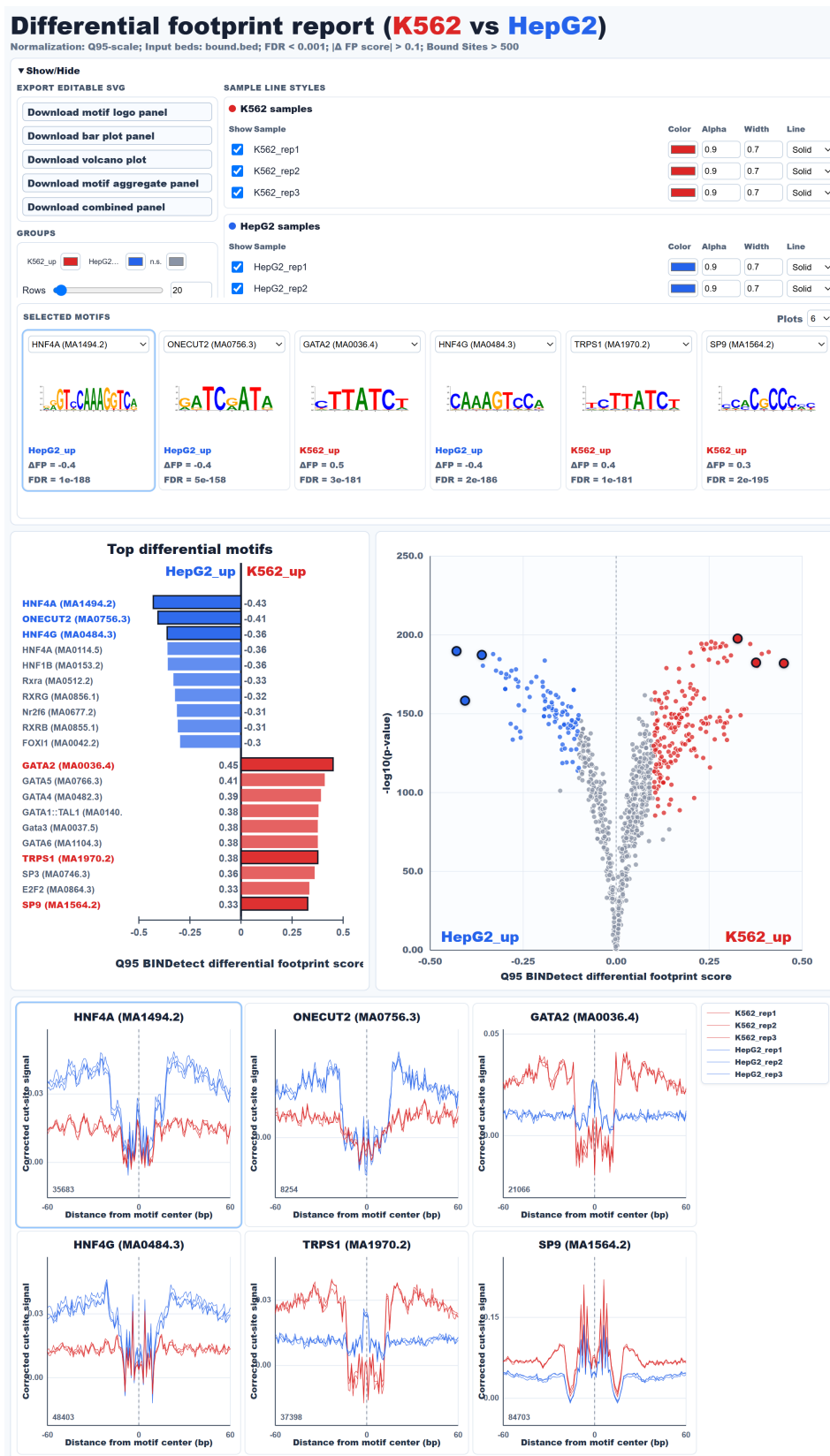


Figure 3. Interactive differential footprint report for K562 versus HepG2. The report summarizes selected differential motifs, motif logos, top differential footprint-score bar plots, a volcano plot, and replicate-level motif-centered aggregate profiles. Blue indicates HepG2-biased footprint scores and red indicates K562-biased footprint scores.

The UMAP panels in Figure 4B show the same regulatory programs in the single-cell embedding. The broad-cell reference panel separates B cells, monocytes, and T/NK cells. STAT6 and FOSB localize primarily to the B-cell region, consistent with IL-4-responsive and AP-1-associated activation programs. CEBPA, IRF8, and RELA localize to the monocyte region, consistent with myeloid differentiation, interferon-regulatory, and NF- κ B inflammatory programs. ZNF683, NR4A1, and SMAD3 localize within the T/NK region, consistent with cytotoxic or resident-like, TCR-response or tolerance-like, and TGF- β -associated regulatory-state programs.

3.5. Workflow Reproducibility, Benchmarking, and GUI Access

The manuscript analyses are reproducible from public input manifests, recorded command settings, and figure-generation scripts. Each analysis writes the resolved input files, software versions, parameters, and plotting tables beside the generated outputs. These records connect the bulk, de novo, pseudobulk, replicate-aware, and report examples back to the exact sample, motif, region, and signal-track choices used for each figure.

The package includes automated checks that run core commands on compact reference inputs and compare key summary outputs for footprint calling, differential footprint analysis, and aggregate plotting. Additional checks exercise public command-line entry points and saved YAML configurations. These tests help ensure that updates preserve expected workflow behavior and shared input/output conventions.

We also benchmarked runtime and peak memory for the example workflow using the same input bundle and matched command outputs for TOBIAS and fp-tools. fp-tools completed the evaluated workflow in 16.7 min, compared with 30.3 min for TOBIAS (Figure 5A), and reduced peak memory from 4.8 GB to 2.7 GB (Figure 5B). These measurements support improved practical efficiency on the evaluated example dataset, while remaining a single-machine benchmark rather than an exhaustive cross-tool performance study.

In addition to command-line execution, fp-tools provides a browser GUI for configuring and running complete footprinting workflows. The GUI exposes the major commands in the sidebar, including atac-correct, call-footprints, match-motifs, diff-footprints, aggregate plotting, and pseudobulk single-cell ATAC workflows. Users can choose a command, load an example, fill in input paths and parameters, review the YAML configuration, launch the run, and then inspect logs, tables, bigWigs, and HTML reports from the run history (Figure 6). This design allows users to complete the same analyses without writing command lines, while the saved YAML configuration remains available for reruns and batch execution.

4. Discussion

fp-tools was developed to make motif-centered ATAC-seq footprinting more reproducible, inspectable, and accessible without changing the core biological interpretation of bias-corrected center-versus-flank footprint signals. Rather than treating bias correction, motif scanning, differential analysis, de novo motif discovery, pseudobulk aggregation, and report generation as separate handoffs, fp-tools organizes them as one command-compatible workflow (Table 1). This design is useful because footprinting analyses often depend on small but important choices in region definitions, motif sets, sample labels, signal scaling, and plotting inputs. Keeping these choices in one recorded workflow makes the analysis easier to review and reproduce.

TOBIAS remains the closest methodological reference point, because it established a widely used and interpretable workflow for ATAC-seq bias correction, footprint scoring, motif-aware differential analysis, and aggregate visualization. fp-tools builds on this signal model while adding reusable YAML configurations, de novo motif discovery

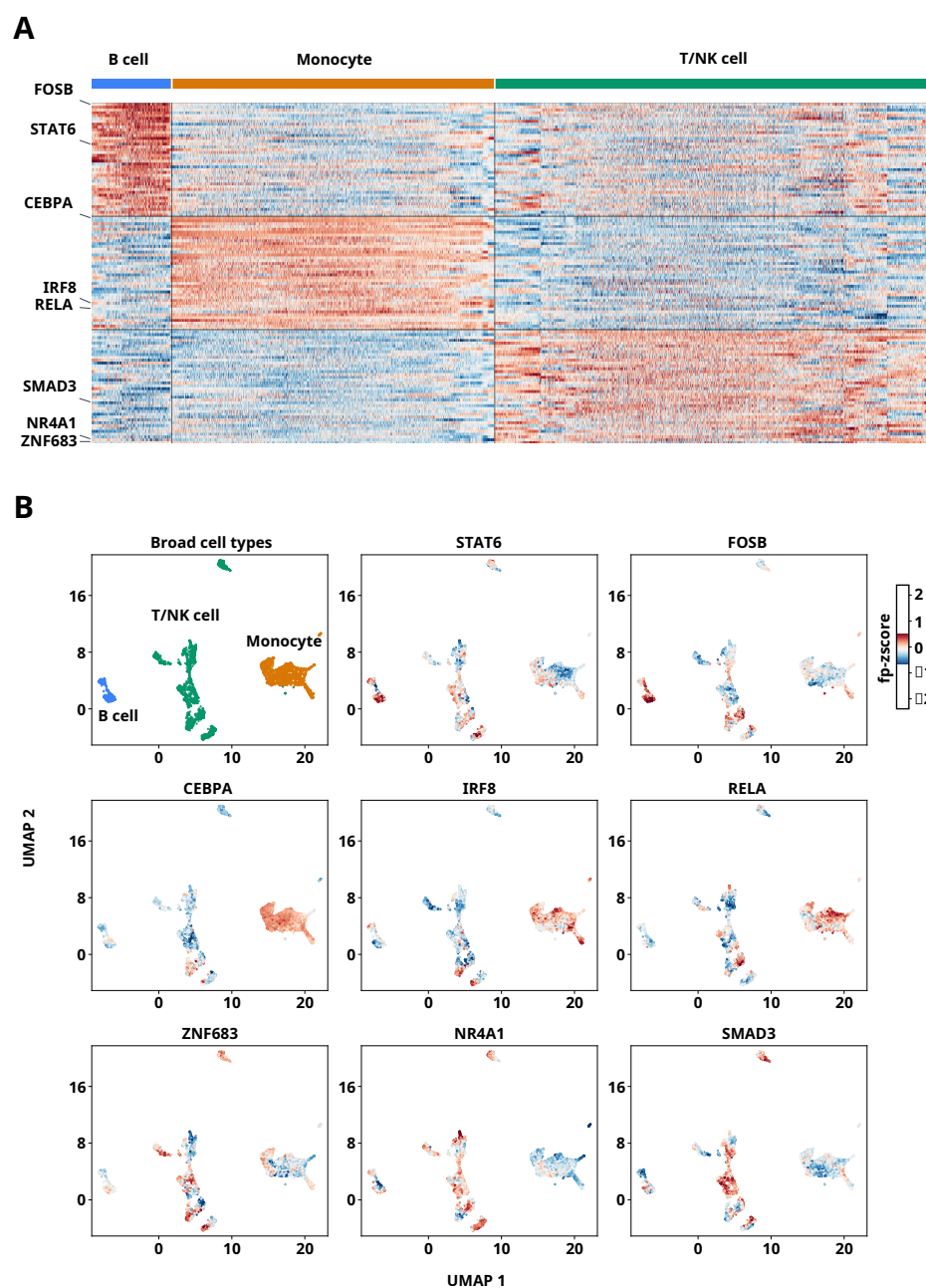


Figure 4. Single-cell footprinting summary in public PBMC5k single-cell ATAC data. **(A)** KNN-smoothed per-cell footprint-signature heatmap for top cell-type-specific motif signatures across B cells, monocytes, and T/NK cells; the TF rows used for Panel B are labelled on the left. **(B)** UMAP projections show broad cell-type annotations and representative motif-associated footprint-signature z-scores. B-cell-associated signatures include STAT6, consistent with IL-4-responsive humoral and class-switch-related signaling, and FOSB/AP-1, consistent with an activation-responsive immediate-early regulatory program. Monocyte-associated signatures include CEBPA, consistent with a C/EBP-family myeloid differentiation program; IRF8, consistent with monocyte/dendritic-cell and interferon-regulatory activity; and RELA, consistent with NF- κ B-mediated inflammatory activation. T/NK-associated regulatory-state signatures include ZNF683/Hobit, consistent with a cytotoxic or resident-like program; NR4A1/Nur77, consistent with TCR-response and tolerance-like regulation; and SMAD3, consistent with TGF- β regulatory signaling. TF labels indicate motif-associated footprint signatures rather than direct TF-specific occupancy.

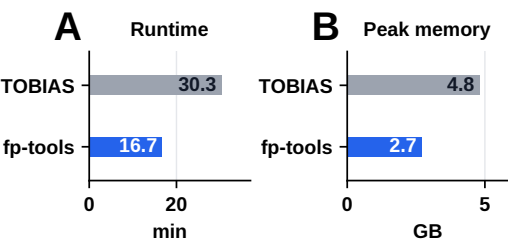


Figure 5. Single-machine runtime and memory benchmark. (A) Runtime and (B) peak memory are summarized for one local analysis run comparing TOBIAS and fp-tools on the same analysis bundle. Lower values indicate faster execution and lower memory use.

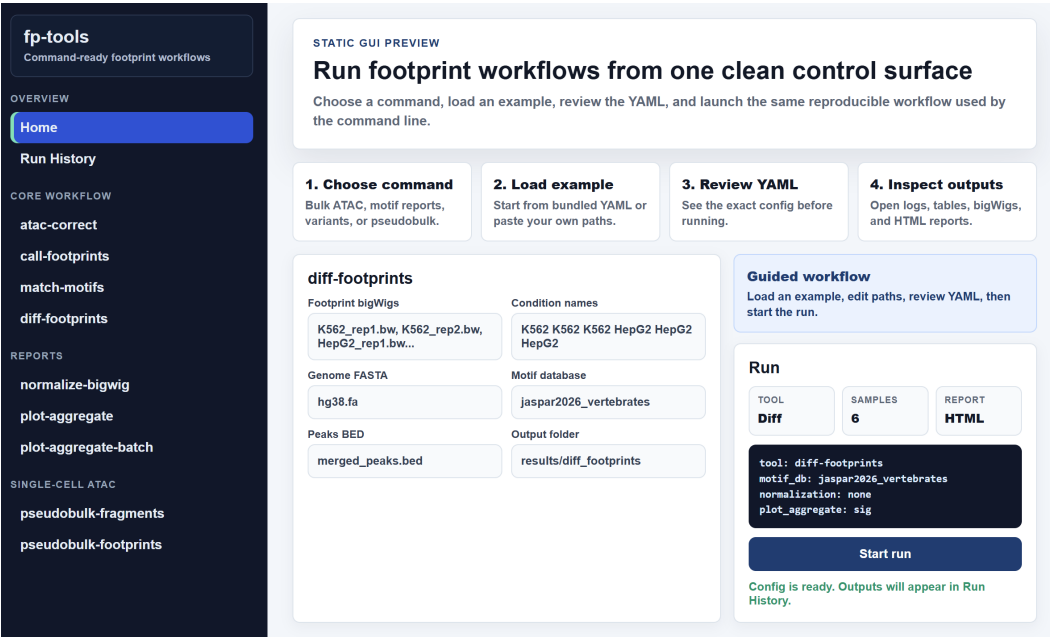


Figure 6. Browser GUI for complete footprinting workflows. The GUI provides sidebar access to core bulk ATAC, report, and pseudobulk commands; guided configuration panels; example loading; YAML preview; run history; and direct review of logs, tables, bigWigs, and HTML reports.

with database matching, replicate-aware reporting, pseudobulk single-cell footprinting, interactive HTML/SVG outputs, a browser GUI, and tested public-data examples. These additions are intended to lower practical barriers for users who need a workflow that can be run from the command line, reviewed in a browser, reproduced from saved configuration files, and traced from public inputs to final figures.

Table 1. Qualitative scope comparison of ATAC-seq regulatory-analysis tools. ✓, native support; ○, partial or adjacent support; NA, outside the tool’s main scope. This table summarizes documented features, not benchmark performance.

Feature	fp-tools	TOBIAS	HINT-ATAC	PRINT/scPrinter	ChromBPNet	max ATAC
Bulk ATAC footprinting	✓	✓	✓	✓	○	NA
Tn5 bias correction	✓	✓	✓	✓	✓	NA
Footprint calling	✓	✓	✓	✓	NA	NA
Motif-aware differential analysis	✓	✓	○	○	NA	NA
De novo motif discovery	✓	NA	NA	✓	○	NA
scATAC/pseudobulk support	✓	○	○	✓	○	○
Replicate-aware reporting	✓	○	○	○	NA	NA
Visualization/reporting	✓	✓	○	✓	✓	○
YAML/batch execution	✓	NA	NA	NA	NA	NA

The public examples illustrate how these additions extend routine footprint analysis. In the ENCODE bulk ATAC-seq examples, de novo motif discovery recovered candidate-derived sequence patterns that produced motif-centered aggregate footprint profiles and overlapped substantially with JASPAR-supported bound-site windows. In the K562/HepG2 replicate comparison, the report combined native motif-background footprint summaries with replicate-supported motif-score statistics and replicate-level aggregate profiles. In the PBMC5k single-cell example, pseudobulk aggregation produced bulk-like inputs for corrected footprint tracks, whereas KNN-smoothed cell-level signatures provided a visualization of cell-type-associated motif-centered cut-site patterns on the UMAP.

These use cases also define the intended interpretation of the reported results. The de novo motif workflow is designed to recover recurring sequence patterns from candidate footprint intervals and to connect those motifs back to motif-centered cut-site structure, rather than to assign TF identity without database matching or additional evidence. The replicate-aware workflow is designed to preserve the familiar motif-versus-background footprint interpretation while adding replicate-level summaries and moderated motif-score statistics. The single-cell workflow is designed to generate reproducible pseudobulk footprint tracks and cell-level visualization signatures, rather than to make direct TF-occupancy calls in individual cells. Together, these examples position *fp-tools* as an auditable route from ATAC-seq cut-site signal to motif-level footprint interpretation.

Several limitations remain. The runtime and memory benchmark was performed on the defined example workflow and demonstrates the efficiency gains in that setting; additional datasets and hardware environments will help quantify scaling across broader use cases. The de novo motif analysis provides operational motif-window recovery summaries rather than formal differential binding significance tests. The cell-level footprint signatures are KNN-smoothed visual summaries of local motif-centered cut-site patterns and should not be interpreted as direct TF-specific occupancy calls in individual cells. Future work will therefore focus on larger controlled benchmarks, broader biological validation, and additional replicate designs across tissues, perturbations, and sequencing depths.

5. Conclusions

We presented *fp-tools* as a Python platform for reproducible, user-accessible ATAC-seq footprint analysis across bulk and single-cell-derived pseudobulk data. The package combines command-line execution, reusable YAML configuration, de novo motif discovery, replicate-aware reporting, pseudobulk footprinting, interactive reports, and a browser GUI in a single workflow. Across public bulk and single-cell examples, *fp-tools* preserved the core interpretation of bias-corrected center-versus-flank footprints while extending the workflow to motif discovery, replicate-supported review, and single-cell pseudobulk analysis.

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Informed Consent Statement: Not applicable.

Data Availability Statement: fp-tools is openly available in the public GitHub repository [oncologylab/fp-tools](#). The repository contains the manifests, schemas, analysis scripts, figure generators, source tables, and small result tables used in this manuscript. Public input data are obtained from the ENCODE Project [19], the 10x Genomics PBMC5k single-cell ATAC example [23], and motif databases such as JASPAR [12] using the included download and preparation scripts. The computational environment is documented in `environment.yml` and the repository Dockerfile.

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Conflicts of Interest: The authors declare no conflicts of interest.

Appendix

Table S1. Primary analysis parameters used for the public ATAC and pseudobulk workflows. The complete source table is provided in the Supplementary Materials as “Primary analysis parameters”.

Step	Parameters
Public input data	Bulk ATAC analyses use public ENCODE K562/HepG2 replicate BAMs and tracks. The pseudobulk single-cell demonstration uses the public 10x Genomics 5k PBMC scATAC fragment dataset. Motif analyses use JASPAR 2026 CORE vertebrate non-redundant motifs.
ENCODE BAM processing	Released GRCh38 alignment BAMs for K562 ENCSR868FGK and HepG2 ENCSR291GJU were filtered to standard chromosomes, indexed, and summarized before footprint analysis.
Blacklist and peak set	Mitochondrial and nonstandard contigs excluded; hg38 blacklist v2 regions subtracted; released ENCODE peak intervals sorted and merged across replicates into one analysis peak set.
Bias correction and footprint scores	<code>atac-correct</code> run on filtered BAMs with merged peaks and blacklist regions using default Tn5-bias settings; corrected cut-site tracks q95-scaled over shared background regions for multi-sample analyses; <code>call-footprints -score footprint</code> run with default footprint and flank windows.
Differential footprint analysis	The 3-vs-3 ENCODE K562/HepG2 replicate footprint tracks were analyzed with repeated condition labels, JASPAR 2026 motifs, and replicate reporting enabled. Single-sample contrasts use motif-site versus background-distribution comparison; replicate-supported analyses use empirical-Bayes variance moderation on per-sample aggregated motif scores.
De novo motif validation	ENCODE K562/HepG2 candidate footprints exported as cell-line-level FASTA files with ± 75 bp flanks; reciprocal STREME 5.5.9 runs used <code>-dna -nmotifs 250</code> ; Tomtom 5.5.9 matched discovered motifs to JASPAR 2026; discovered motifs were scanned back into candidate intervals for motif-window overlap summaries and aggregate footprint review.
Pseudobulk PBMC analysis	Public 10x PBMC5k scATAC fragments grouped into broad B-cell, monocyte, and T-NK labels, retaining 4,437 annotated cells and requiring at least 300 cells and 50,000 fragments per group. The full <code>pseudobulk-footprints</code> route writes pseudo-BAMs, runs ATAC correction with zero read shifting for fragment-derived pseudobulk BAMs, and emits footprint-score tracks, motif-aware differential results, marker heatmaps, and KNN-smoothed footprint-signature UMAPs.

Table S2. Software and resource versions used for manuscript analyses. The complete source table is provided in the Supplementary Materials as “Software and resource versions”.

Tool or resource	Version	Role
bedtools	2.31.1	Region sorting, merging, and blacklist subtraction
Biopython	1.87	Sequence and motif-related utilities
Cell Ranger ATAC	2.0.0	Source processing for the public 10x PBMC5k scATAC dataset
fp-tools	0.1.7	ATAC correction, footprint calling, motif matching, differential footprint analysis, pseudobulk utilities, and plotting helpers
hg38 blacklist	v2	Artifact-region filtering for processed bulk ATAC alignments
hg38 reference genome	GRCh38/hg38	Sequence extraction and ATAC bias correction
htslib	1.23.1	SAM/BAM/VCF I/O backend used by samtools
JASPAR CORE vertebrates	2026 non-redundant	Known motif database for differential footprint analysis
matplotlib	3.10.9	Manuscript figure rendering
MEME Suite, STREME, and Tomtom	5.5.9	De novo motif discovery and discovered-motif matching
NumPy	2.4.6	Numerical arrays and benchmark/figure calculations
pandas	2.3.3	Tabular processing for benchmark, manifest, and figure builders
pyBigWig	0.3.25	bigWig signal reading and writing
pysam	0.23.3	BAM and genomic interval I/O
Python	3.12.3	Runtime for analyses, figure builders, and provenance extraction
samtools	1.23.1	Released ENCODE BAM filtering, indexing, and QC summaries
SciPy	1.17.1	Statistical and interpolation utilities

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